

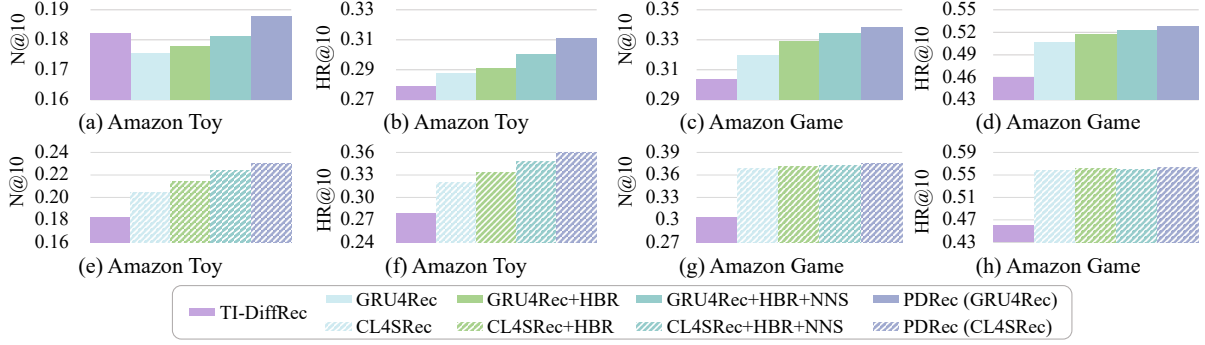


Figure 5: Results of PDRec on GRU4Rec/CL4SRec and their ablation versions on Toy and Game datasets.

Setting	Algorithms	N@1	N@5	N@10	N@20	N@50	HR@5	HR@10	HR@20	HR@50	AUC
Game	T-DiffRec (M)	0.0981	0.1520	0.1727	0.1934	0.2375	0.2029	0.2673	0.3494	0.5780	0.5924
	TI-DiffRec (M)	0.1053	0.1598	0.1806	0.2008	0.2407	0.2111	0.2759	0.3562	0.5623	0.5932
	SASRec (M)	0.1267	0.2019	0.2261	0.2490	0.2785	0.2722	0.3472	0.4380	0.5873	0.5951
	+HBR	0.1283	0.2061	0.2311	0.2533	0.2835	0.2785	0.3558	0.4438	0.5972	0.6092
	+HBR+NNS	0.1264	0.2068	0.2323	0.2542	0.2844	0.2815	0.3606	0.4480	0.6013	0.6123
	+HBR+NNS+DPA	0.1302	0.2093	0.2348	0.2574	0.2873	0.2826	0.3616	0.4515	0.6026	0.6106
Toy	T-DiffRec (M)	0.1674	0.2643	0.2977	0.3247	0.3597	0.3548	0.4584	0.5655	0.7428	0.7232
	TI-DiffRec (M)	0.1709	0.2757	0.3096	0.3378	0.3721	0.3723	0.4773	0.5887	0.7622	0.7407
	SASRec (M)	0.2273	0.3532	0.3905	0.4190	0.4467	0.4674	0.5826	0.6955	0.8342	0.8007
	+HBR	0.2332	0.3597	0.3963	0.4250	0.4547	0.4741	0.5872	0.7006	0.8501	0.8145
	+HBR+NNS	0.2352	0.3601	0.3975	0.4257	0.4557	0.4733	0.5890	0.7002	0.8517	0.8138
	+HBR+NNS+DPA	0.2363	0.3623	0.3992	0.4275	0.4572	0.4761	0.5904	0.7022	0.8520	0.8153

Table 3: Ablation versions of PDRec on two CDSR datasets. All improvements are significant compared to baselines.

vancements in SR in the future, thereby extending the lifespan of the proposed DM-utilization frameworks.

(2) Progressive improvements are discernible among distinct versions of PDRec, with PDRec outperforming all its variants. It demonstrates that the proposed components are effective and universal for different base sequential encoders and datasets, further

5.5 Results of Cross-domain SR (RQ4)

PDRec could also benefit positive transfer in CDSR. We follow typical CDSR settings (??) and employ PDRec with SASRec (M) (M indicates directly mixing both source and target domains’ behaviors in chronological order) on Toy→Game and Game→Toy settings. We also implement T-DiffRec (M) and TI-DiffRec (M)

(1) PDRec outperforms all diffusion-based models in CDSR, which implies that PDRec could be used in other tasks such as cross-domain scenarios. Its HBR provides an intuitive but effective way to filter negative transfers in cross-domain recommendation (i.e., mixing all domains’ behaviors sequentially and conducting reweighting via diffusion), which could be

(2) PDRec outperforms all of its ablation versions on most CDSR settings, with each component contributing to incremental improvements. It reconfirms the effectiveness and universality of HBR, NNS, and DPA from diffusion model.

(3) It is impressive that PDRec exhibits notable improve-

ments across various metrics compared to the original T-DiffRec/TI-DiffRec on the mixed behavior sequence (**up to 38.3%**). It reiterates our main contribution that takes full advantage of the outputs of diffusion model as a plugin in SR.

6 Conclusion

In this paper, we propose an effective and model-agnostic Plug-in Diffusion Model for Recommendation (PDRec) framework. Instead of focusing on the highest-score item, PDRec fully leverages the diffusion-based preferences on all items. PDRec employs a historical behavior reweighting method to identify the indispensable behaviors and conducts a knowledge extracting strategy from both the unobserved items via the diffusion-based positive augmentation, and noise-free negative sampling. The extensive experiments and analyses on four datasets, three base models and two recommendation tasks demonstrate the effectiveness and universality of PDRec. In the future, we will continue to explore the tailored hard negative sampling strategies in PDRec and attempt to adapt PDRec as a flexible and detachable

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